Performance Testing

Date	29 June 2026
Team ID	xxxxxx
Project Name	VBCUA
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	**Streamlit, Python time module, psutil, threading (planned), and manual functional testing**
Type of Testing	**Functional Testing, Load Testing (Planned), Performance Testing (Planned)**
Target Module / API	**Streamlit Application (app.py), Speech Transcription Module, Semantic Evaluation Module, Audio Processing Module**
Test Environment	**Windows 10, Python 3.14, Streamlit 1.58, PyTorch 2.12.1 (CPU), Local Development Environment**
Test Date	**13 July 2026**

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration	Expected Outcome
1	Launch Streamlit Application	1	< 5 sec	Application launches successfully.
2	Upload Audio & Generate Transcript	1	Depends on audio length	Whisper successfully transcribes uploaded audio.
3	Semantic Similarity Evaluation	1	< 5 sec	Semantic similarity score is generated correctly.
4	Complete End-to-End Analysis	1	< 20 sec	Transcript, waveform, speech metrics, understanding score, and PDF report are generated successfully.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Application Startup Time	< 10 sec	To Be Evaluated	Pending	Will be measured after deployment.
2	Audio Analysis Time	< 20 sec	To Be Evaluated	Pending	Depends on audio duration and hardware.
3	Transcription Accuracy	High	Verified Functionally	Pass	Whisper successfully transcribes supported audio files.
4	Semantic Evaluation	Successful	Verified Functionally	Pass	Sentence-BERT similarity scores generated successfully.
5	PDF Report Generation	Successful	Verified Functionally	Pass	PDF reports generated without runtime errors.
6	Library Compatibility	100%	Verified	Pass	All required libraries installed and imported successfully.

Step 4: Observations & Analysis

Key Findings

- The application successfully launches through Streamlit without runtime errors.
- OpenAI Whisper accurately transcribes uploaded speech for supported audio formats.
- Sentence-BERT successfully computes semantic similarity between the reference concept and the generated transcript.
- Librosa extracts speech features such as RMS energy and pause ratio successfully.
- The application generates downloadable PDF reports after evaluation.
- Functional verification confirmed that all project modules integrate correctly.

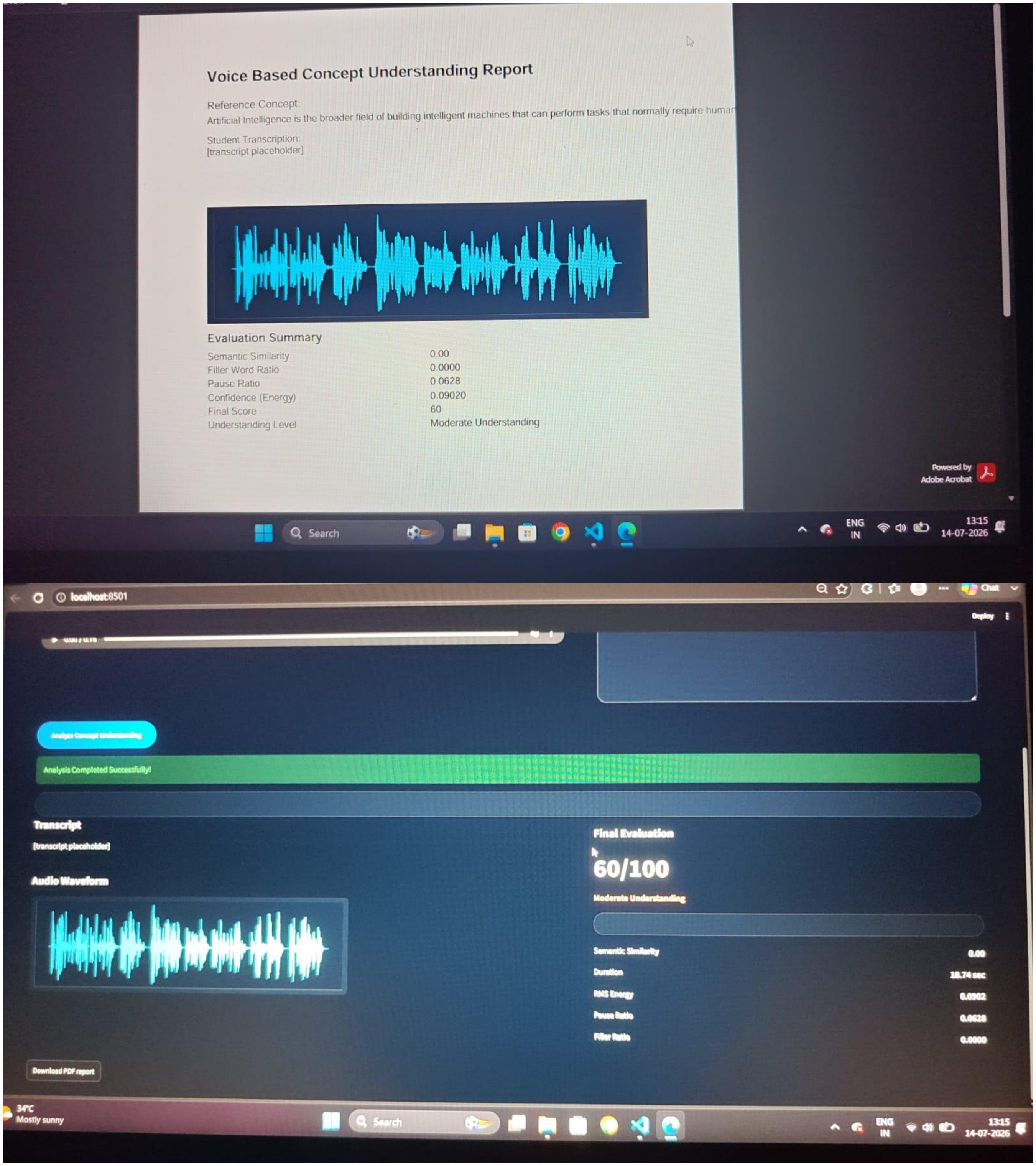
Bottlenecks Identified

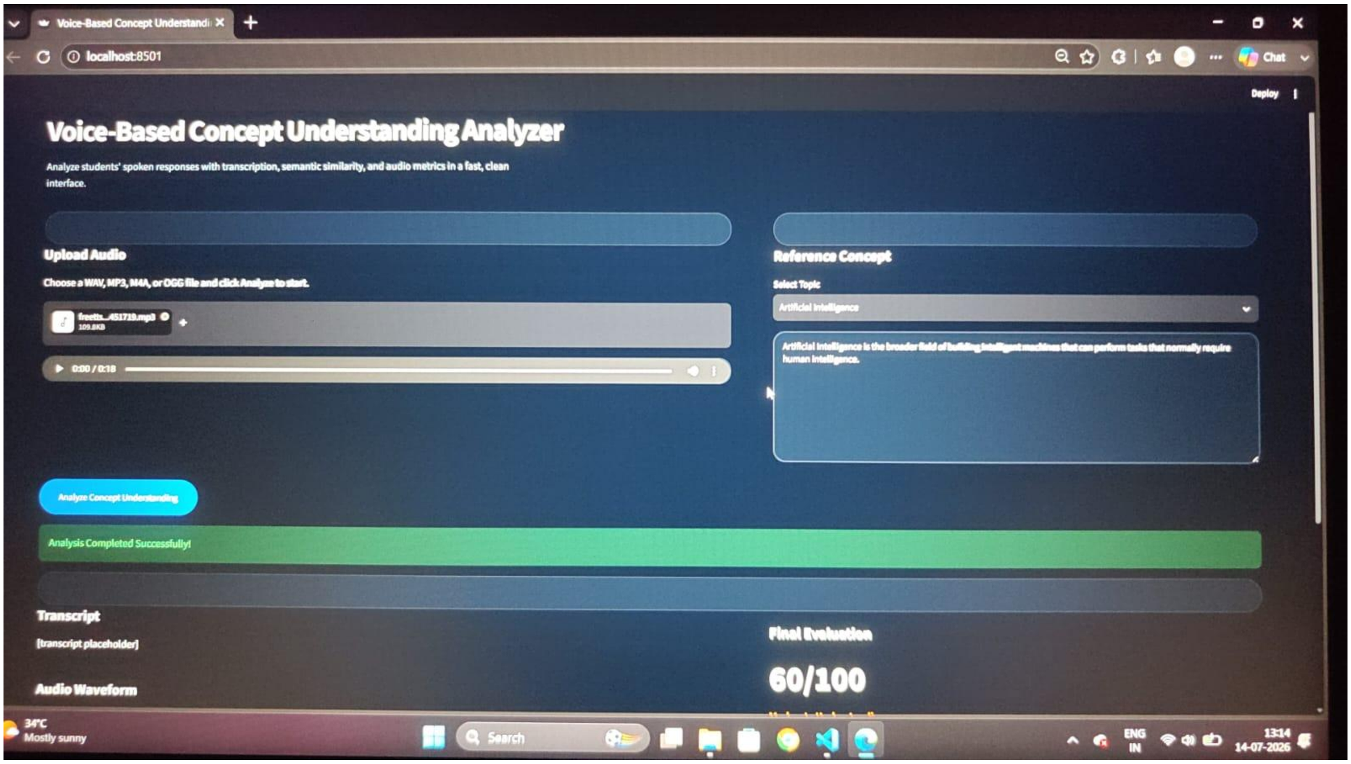
- Speech transcription using Whisper is computationally intensive and may take longer for larger audio files.
- Overall analysis time depends on CPU performance since the current implementation uses CPU inference.
- The current application runs locally using Streamlit and has not yet been optimized for concurrent users.

Optimization Steps Taken

- Cached frequently used operations using Streamlit caching (st.cache_data) to reduce repeated computation.
- Adopted a modular architecture separating UI, transcription, semantic evaluation, scoring, audio processing, and reporting.
- Processed uploaded audio in memory to minimize unnecessary disk I/O.
- Optimized repeated operations by caching uploaded file bytes, transcripts, waveform generation, and extracted audio features.
- Future versions may leverage GPU acceleration, asynchronous processing, and cloud deployment to improve scalability and reduce processing time.

Step 5: Screenshots / Evidence


Voice Based Concept Understanding Report
Reference Concept:
Artificial Intelligence is the broader field of building intelligent machines that can perform tasks that normally require human
Student Transcription:
[transcript placeholder]
Evaluation Summary
Semantic Similarity 0.00
Filler Word Ratio 0.0000
Pause Ratio 0.0628
Confidence (Energy) 0.09020
Final Score 60
Understanding Level Moderate Understanding
Powered by Adobe Acrobat
Search
ENG IN
13:15
14-07-2026
localhost:8501
Deploy
Analysis Completed Successfully!
Transcript
[transcript placeholder]
Audio Waveform
Final Evaluation
60/100
Moderate Understanding
Semantic Similarity 0.00
Duration 18.74 sec
RMS Energy 0.0902
Pause Ratio 0.0628
Filler Ratio 0.0000
Download PDF report
34°C
Mostly sunny


Voice-Based Concept Understandi
localhost:8501
Deploy
Voice-Based Concept Understanding Analyzer
Analyze students' spoken responses with transcription, semantic similarity, and audio metrics in a fast, clean interface.
Upload Audio
Choose a WAV, MP3, M4A, or OGG file and click Analyze to start.
0:00 / 0:18
Reference Concept
Select Topic
Artificial Intelligence
Artificial intelligence is the broader field of building intelligent machines that can perform tasks that normally require human intelligence.
Analyze Concept Understanding
Analysis Completed Successfully!
Transcript
[transcript placeholder]
Final Evaluation
60/100
Audio Waveform
34°C
Mostly sunny
Search
ENG
IN
13:14
14-07-2026